Build your own J

A phone call that runs your laptop. Built on Sarvam AI by Hari and Anurag at the Sarvam AI hackathon, in collaboration with IIT Hyderabad.

```
Your phone → Sarvam phone number → Sarvam voice agent "J"
  (Saaras listens · Sarvam-105B decides · Bulbul speaks)
→ secure tunnel → the ARRC server on your laptop → your laptop does it
```

You call a number, talk to J, and your laptop does the work. J can open sites and apps, find and open files, play music, take notes, set timers, lock the screen, draft emails, and upload assignments to Google Classroom. She can also call **you** when a deadline is close.

> **Easiest way: let Claude Code do it (about 20 minutes).** Unzip the folder, install Claude Code (the desktop app) from `claude.com/claude-code`, open the `ARRC-Build-Your-Own-J` folder in it and type **Set up J for me**. Claude installs everything on your laptop and tells you exactly what to click on the Sarvam website, with each value already on your clipboard. No coding needed. The rest of this guide is the same steps by hand.

Without Claude Code it takes about 30 minutes. No typing commands: you double-click two files and copy-paste into the Sarvam website.

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0. What you need

- **A Windows laptop.** The laptop actions use Windows tools.
- **A Sarvam account** at `indus.sarvam.ai` with some credits. Calls use credits.
- **A phone number in Sarvam.** Get it under **Deploy** → **Phone Numbers**. This is the number you call.
- **Google Chrome**, only if you want the Classroom uploads.

Node.js and cloudflared (a free tool that connects J to your phone) install themselves in step 1.

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1. Switch J on 5 min

1. Unzip `ARRC-Build-Your-Own-J.zip` somewhere easy, like your Desktop, and open the folder.
2. Double-click **Start-J.bat**. If Windows asks "Do you want to allow this app...", click **Yes**. The first time, it installs what it needs (a few minutes).
3. It asks for your mobile number. Type it with the country code, numbers only, like `919876543210`, and press Enter. Calls from any other phone are refused.
4. Wait until it says **J is running**. Keep that window open. Closing it switches J off.

5. Open `http://localhost:3710` in your browser. That's the ARRC dashboard, where everything J does shows up live.

> **The copy helper.** Double-click **Copy-for-Sarvam.bat**. When a step says **copy N**, press that number and Enter there, then press **Ctrl+V** on the Sarvam website. 1 = J's instructions, 2 = greeting text, 3 = tool description, 4 = tool URL, 5 = J's password, 6 = lock J to your phone.

> **Never share the file arrc\.env.** It holds J's password and, later, your Sarvam key.

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2. Create J in Sarvam 10 min

Go to `indus.sarvam.ai` → **Build** → **Agents** → **Create from scratch**. Name it **J**.

1. **Instructions:** copy 1, paste.
2. **Variables:** add an input variable called `opening_line`. For its default value, copy 2, paste. It is:

   ```
   Hey Boss, it's J. What can I do for you?
   ```

3. **Greeting:** set it to `{{opening_line}}`. If Sarvam asks for greeting translations, use `{{opening_line}}` for those too. This lets J open her own calls with the reason she's calling.
4. **Settings:**
   - Voice: pick one you like. We used **Simran**.
   - Starting language: **English**. Switch language during call: **on**, so she follows you into Hindi, Telugu and more.
   - Turn **off** "Hang up after unanswered nudges". Otherwise she cuts the call while you look at your laptop.
   - Background sound: none.

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3. Give J her one tool 10 min

In the agent, go to **Tools** → **Add tool** and make an **API / HTTP tool**:

Field	What to put
Name	`laptop_action`
Description	copy 3, paste
When it runs	During call
Method	POST
URL	copy 4, paste. It looks like `https://something-random.trycloudflare.com/action`
Auth	Bearer token. Make a new secret: copy 5, paste. Pick it.
Parameters	`action` (text, filled by the agent) `input` (text, filled by the agent) `caller` (pick **User Identifier**, so ARRC knows it's really you calling)
Max wait	10 seconds

Now test it: press **Test agent** in Sarvam and say "what's my battery?". The answer should also appear on your dashboard at `localhost:3710`.

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4. Put J on your phone number 5 min

1. **Commit** the agent. Sarvam saves it as a version, like v1.
2. Go to **Deploy** → **Phone Numbers**, pick your number, and set **Inbound** to agent J and the version you just committed.
3. Call your Sarvam number from your own phone.
4. It works? In Copy-for-Sarvam.bat press **6**. Now J only listens to your phone. (The Test agent button stops working after this; that's expected.)

Things to say

- "Play Arijit Singh on YouTube"
- "Open my resume"
- "Note this down: call the bank tomorrow at 10"
- "Set a 5 minute timer for tea"
- "Draft an email to rahul@example.com saying the demo is at 10" (she opens a Gmail draft, you press send)
- "How much battery do I have?"
- "Lock my laptop" (she asks first)

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5. Optional: J calls you

J can phone you before a deadline, and when your battery is about to die. Say "remind me, maths project due tomorrow 5 pm" on a call, or add a deadline on the dashboard.

1. In Sarvam, go to **Settings** and create a **workspace API key**. Copy your **Org ID** and **Workspace ID** from there too.
2. Fill these in `.env`:
   - `SARVAM_WORKSPACE_KEY`, `SARVAM_ORG_ID`, `SARVAM_WORKSPACE_ID`: from step 1.
   - `ARRC_APP_ID`: the agent's ID. It's the last part of the agent's address when you edit it in Sarvam.
   - `ARRC_APP_VERSION`: the version number you deployed, like `1`.
   - `ARRC_AGENT_NUMBER` and `ARRC_CONNECTION_ID`: your Sarvam number and its connection, under Deploy → Phone Numbers.
3. Save the file and press **Test urgent call** on the dashboard. Your phone should ring.

> Every time you commit a new version of J, update `ARRC_APP_VERSION` and save, or J calls you with the old version.

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6. Optional: upload assignments to Google Classroom

1. In `.env` set `CLASSROOM_SUBMIT=attach` and save.
2. A separate Chrome window opens. Sign in to your college Google account there, once. Leave that window open.
3. On a call, say "J, my interview report is done, upload it". She finds the assignment and the matching file on your laptop and attaches it.

> **attach** only attaches the file. You still check it and press **Turn in** yourself. That's on purpose. `live` turns work in automatically; use it only if you're sure.

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7. Add your own tool

This is the fun part. Every action is a few lines in `arrc/server.js`, inside `const ACTIONS = {`. Add your own next to the others, like this one that opens a GitHub profile:

```
async open_github({ input }) {
  const user = clean(input);
  if (!user) return 'Whose GitHub should I open?';
  await openUrl(`https://github.com/${encodeURIComponent(user)}`);
  return `Opened GitHub for ${user}.`;
},
```

1. Save `server.js`. The J window restarts the server by itself.
2. Add `open_github` to the list of actions in the tool description (step 3) and add one line about it in J's instructions.
3. Commit and deploy the new version. Now say "open GitHub for torvalds".

Whatever your action returns is what J says back to you. Keep it short.

8. If something goes wrong

What happens	Fix
J says she couldn't reach your laptop	Start-J.bat isn't open, or the tool URL changed. Double-click Start-J.bat; if it says CHANGED, copy 4 into the tool URL in Sarvam, commit and deploy.
"Sorry, I can only take orders from my owner's phone"	Check `OWNER_NUMBER` in `.env`: digits only, with country code. Save it; the server restarts by itself.
She starts in Hindi	Settings → Starting language → English.
She hangs up on her own	Turn off "Hang up after unanswered nudges".
Her outbound calls use old instructions	Update `ARRC_APP_VERSION` in `.env` and save.
Classroom: "the window is signed out"	Sign in again in the separate Chrome window ARRC opened.

Keep it safe

- Only your phone number is obeyed (`OWNER_NUMBER`), and every tool call needs your password.
- J can only do the actions listed in `server.js`. Nothing you say on a call runs as a command.
- Don't post your `.env`, your tunnel address or your Sarvam keys anywhere.

ARRC by Hari & Anurag. Built at the Sarvam AI hackathon, in collaboration with IIT Hyderabad. Go build your own J.